Probability II

Anurag (bmat2508@isibang.ac.in)

Instructor: Mathew Joseph

Class 1, January 9 2026

Some notations:

- Ω : Sample Space (for now we will assume that it is finite or countable).
- ω : Outcomes belonging to the sample space.
- Probability $\mathbb{P}(\omega)$ that $0 \geq \mathbb{P}(\omega) \geq 1$ and $\sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1$.
- A Random Variable $X : \Omega \to \mathbb{R}$.

If X and Y are two random variables then their joint PMF
$\mathbb{P}_{X,Y}(x_i, y_j) = \mathbb{P}(X = x_i, Y = y_j) = \mathbb{P}(\{\omega : X(\omega) = x_i, Y(\omega) = y_j\})$

1 Simple Random Walks

$X_1, X_2, \ldots$ is a sequence of i.i.d Ber($\frac{1}{2}$) random variables wherein for each i we have:

$$X_i = \begin{cases} 1 & \text{, w.p } \frac{1}{2} \\ -1 & \text{, w.p } \frac{1}{2} \end{cases}$$

We consider the sequence of partial sums defined as $S_0 = 0$ and $S_n = X_n + S_{n-1}$. Clearly $|S_{i+1} - S_i| = 1$ and knowing either the first n sums or the values taken by the first n random variables would readily determine the other. Each path has an equal probability of either going up or down. We wish to determine $\mathbb{P}(S_n = k)$.
Let us denote $N_n(k)$ to be the number of paths ending at k within the first n steps. Clearly then,

$$\mathbb{P}(S_n = k) = \frac{N_n(k)}{2^n}$$

If we have α many $+1$'s and β many -1's, then clearly:

$$\alpha + \beta = n,$$

$$\alpha - \beta = k$$

Solving these we get $\alpha = \frac{n+k}{2}$ and $\beta = \frac{n-k}{2}$. Thus $N_n(k) = \binom{n}{\frac{n+k}{2}}$. Furthermore, if our paths start at (b, a) and end at (b', a') then the number of possible paths is

$$\binom{b'-b}{\frac{b'-b+a'-a}{2}}$$

Now we wish to find $\mathbb{P}(S_1 > 0, S_2 > 0, \ldots, S_n = k)$. Let $N_n^+(k)$ denote the number of paths staying above the x-axis and reaching k at the n-th step.
We consider the number of paths from (a, b) to (a', b') that touch the x-axis. By the **Reflection Principle** we have that these paths map bijectively to those that go from $(a, -b)$ to (a', b') with the part prior to the point of first touching the x-axis reflected. We also require that our first step is a $+1$.
Then our required probability is:

$$\mathbb{P}(S_1 = 1, S_n = k) - \mathbb{P}(S_1 = 1, S_n = k, S_i = 0 \text{ for some } i \geq 2)$$

The first probability is just

$$\mathbb{P}(X_1 = 1)\mathbb{P}(X_2 + \cdots + X_n = k - 1) = \frac{\binom{n-1}{\frac{n+k}{2}-1}}{2^n}$$

and the second one is

$$\frac{\binom{n-1}{\frac{n+k}{2}}}{2^n}$$

and thus our required probability is:

$$\frac{k}{n}\frac{\binom{n}{\frac{n+k}{2}}}{2^n} = \frac{k}{n}\mathbb{P}(S_n = k)$$

Now we wish to find the probability that the first return to the x-axis occurs after the $2n$-th step. If this occurs at the $2m$-th step then we must reach 1 at the $(2m-1)$-th step while staying above the x-axis. This can be done in $N_{2m-1}^+(1)$ ways and hence the required probability is:

$$\sum_{m=n}^{\infty} \frac{\mathbb{P}(S_{2m-1} = 1)}{2(2m-1)}$$

which simplifies to

$$\frac{N_{2n-1}(1)}{2^{2n-1}}$$

Let's find the probability that the path never returns to 0. Since this sequence of events is decreasing, it is enough to take the limit of the probability we just found.

$$\mathbb{P}(\text{Path never returns to } 0) = \lim_{n\to\infty} \frac{\binom{2n}{n}}{2^{2n}}$$

We employ Stirling's approximation: $n! \sim \sqrt{2\pi} n^{n+\frac{1}{2}} e^{-n}$, using which we obtain that our required limit is 0. Therefore

$$\mathbb{P}(\text{RW returns to } 0) = 1$$

but it is important that the coin is fair, otherwise our random walk may never return to 0.
Even while performing a random walk in the integer lattice, the probability that we return to the origin is 1. But for higher dimensions, the probability is strictly between 0 and 1. Dimensions 1 and 2 are called recurrent and the higher ones transient.

Class 2, January 14 2026

Let M_n denote the first time our random walk attains its maximum value. We have the following result:

Theorem 1. $\mathbb{P}(an \leq M_n \leq bn) \to \frac{2}{\pi}[\arcsin\sqrt{b} - \arcsin\sqrt{a}]$, *where* $a, b \in [0,1]$.

Limiting CDF of $\frac{M_n}{n}$ is

$$F(t) = \frac{2}{\pi}\arcsin\sqrt{t}$$

and thus, the limiting PDF

$$f(t) = \frac{1}{\pi\sqrt{t(1-t)}}$$

This is known as the Arcsine Law.

Proof. Fix $0 < m < n$. We intend to find the probability $\mathbb{P}(M_n = m)$. This is equivalent to saying $S_m > S_0, \ldots, S_m > S_{m-1}$ and $S_m \geq S_{m+1}, \ldots, S_m \geq S_n$, and thus our probability simplifies to

$$\mathbb{P}(M_n = m) = \mathbb{P}(S_m > S_0, \ldots, S_m > S_{m-1})\mathbb{P}(S_m \geq S_{m+1}, \ldots, S_m \geq S_n)$$

We use the following result we had seen earlier

Lemma 1 (Basic Lemma)**.** $\mathbb{P}(S_1 \neq 0, \ldots, S_{2\nu} \neq 0) = \mathbb{P}(S_{2\nu} = 0)$

and thus $\mathbb{P}(S_1 > 0, \ldots, S_{2\nu} > 0) = \frac{1}{2}\mathbb{P}(S_{2\nu} = 0)$ □

Now we wish to find $\tilde{M}_n$, the last time of attaining maximum. It would be convinient to look at the walk backwards, wherein we would get from the previous result

$$\mathbb{P}(n - \tilde{M}_n \leq an) \to \frac{2}{\pi}\arcsin\sqrt{a}$$

Consider now, a walk of length $n = 2k$. We wish to find the probability that the last visit to the origin occurs at time 2ν. But this is simply

$$\mathbb{P}(S_{2\nu} = 0, S_{2\nu+1} \neq 0, \ldots, S_{2k} \neq 0) = \mathbb{P}(S_{2\nu} = 0)\mathbb{P}(S_{2\nu+1} \neq 0, \ldots, S_{2k} \neq 0)$$

Using the previous notation this is nothing but $u_{2\nu}u_{2k-2\nu}$.

Class 3, January 16 2026

Let f_{2k} denote the probability that the first return to 0 happens at time $2k$. But this can be written as

$$\mathbb{P}(S_1 = 1, S_2 > 0, \ldots, S_{2k-1} = 1, S_{2k} = 0) + \mathbb{P}(S_1 = -1, S_2 < 0, \ldots, S_{2k-1} = -1, S_{2k} = 0)$$

By symmetry, both probabilities are the same and using the Ballot Theorem, we get that the required probability is

$$2\frac{N^+_{2k-1}(1)}{2^{2k}} = \frac{\binom{2k-1}{k}}{(2k-1)2^{2k-1}} = \frac{u_{2k}}{2k-1}$$

Lemma 2.

$$u_{2n} = f_2 u_{2n-2} + \cdots + f_{2n} u_0$$

Proof. This is easy to see. Notice that

$$\mathbb{P}(S_{2n} = 0) = \sum_{k=1}^{n} \mathbb{P}(S_{2n} = 0 \mid \text{first return occurs at time } 2k) f_{2k}$$

wherein we condition on the time of first return to the origin. □

Theorem 2. *The probability that in the time interval 0 to $2n$, SRW spends $2k$ time units on the positive side and $2n - 2k$ time units on the negative side is*

$$\alpha_{2k,2n} = u_{2k} u_{2n-2k}$$

Therefore for $0 < t < 1$,

$$\mathbb{P}(\textit{SRW spends} \leq tn \textit{ time units on the positive side}) \to \frac{2}{\pi} \arcsin \sqrt{t}$$

Proof. We define the following

$$b_{2k,2n} = \mathbb{P}(\text{RW of length } 2n \text{ has } 2k \text{ sides above x-axis})$$

Firstly, we see that $b_{2n,2n} = u_{2n} = \alpha_{2n,2n}$ and that $b_{0,2n} = \mathbb{P}(S_1 \leq 0, \ldots, S_{2n} \leq 0) = u_{2n} = \alpha_{0,2n}$ Assume exactly $2k$ sides out of $2n$ are positive, $1 \leq k \leq n-1$. Say that the first return to 0 is at time $2r < 2n$.
Case 1: SRW spends $2r$ time units in the positive side, clearly then $r \leq k$. And the path from $(2r, 0)$ till time $2n$ has $2k - 2r$ sides above the x-axis. The number of such paths is

$$\frac{1}{2} 2^{2r} f_{2r} (2^{2n-2r} b_{2k-2r,2n-2r})$$

Case 2: SRW spends $2r$ time units on the negative side. As a consequence, we see that $n - r \geq k$ In this case, the number of such paths is

$$\frac{1}{2} 2^{2r} f_{2r} (2^{2n-2r} f_{2k,2n-2r})$$

Thus we have

$$b_{2k,2n} = \frac{1}{2}\sum_{r=1}^{k} f_{2r} b_{2k-2r,2n-2r} + \frac{1}{2}\sum_{r=1}^{n-k} f_{2r} b_{2k,2n-2r}$$

We keep k fixed and proceed by induction. Assume $b_{2k,2n} = \alpha_{2k,2n}$ for $n \leq N-1$.

$$\begin{aligned} b_{2k,2N} &= \frac{1}{2}\sum_{r=1}^{k} f_{2r} u_{2k-2r} u_{2N-2k} + \frac{1}{2}\sum_{k=1}^{n-k} f_{2r} u_{2k} u_{2N-2r-2k} \\ &= \frac{1}{2} u_{2N-2k} \sum_{r=1}^{k} f_{2r} u_{2k-2r} + \frac{1}{2} u_{2k} \sum_{r=1}^{N-k} f_{2r} u_{2N-2r-2k} \\ &= \frac{1}{2} u_{2N-2k} u_{2k} + \frac{1}{2} u_{2k} u_{2N-2k} \\ &= u_{2N-2k,2k} = \alpha_{2k,2N} \end{aligned}$$

□

Class 4, January 21 2026

Generating Functions

Definition 1. *Let $\{a_n\}_{n\in\mathbb{N}}$ be a sequence of real numbers. Then the generating function of this sequence is*

$$A(x) = \sum_{n\geq 0} a_n x^n$$

If the sequence is bounded, then the power series converges. So, $A^{'}(x)$ is defined for all $|x| < 1$. If X is a random variable taking values in $\{0, 1, ...\}$, then the probability generating function for this random variable is

$$P(x) = \sum_{k\geq 0} p_k x^k$$

where $p_k = \mathbb{P}(X = k)$. Clearly, this power series is absolutely convergent. We further define $q_j = \mathbb{P}(X > j)$. Then calculations show that

$$Q(x) = \frac{1 - P(x)}{1 - x}$$

We know from Probability I that $Q(1) = \mathbb{E}[X]$, but we would like to see how this follows from here. We see that

$$P^{'}(x) = \sum_{k=1}^{\infty} k p_k x^{k-1}$$

and thus $P^{'}(1) \to \mathbb{E}[X]$. By the Mean Value Theorem, we have $Q(x) = P^{'}(x_0)$ where $x_0 \in (x, 1)$. So,

$$Q(1) = \lim_{x \to 1^-} Q(x) = \lim_{x \to 1^-} P^{'}(x) = \mathbb{E}[X]$$

We could also compute the variance in terms of the generating function.
Exercise: Find the variance in terms of the generating function.
If we have two independent nonnegative integer valued random variables X and Y, we have

$$P_{X+Y}(s) = \mathbb{E}[s^{X+Y}] = \mathbb{E}[s^X]\mathbb{E}[s^Y] = P_X(s)P_Y(s)$$

With this, we have that the coefficient of s^{m+n} in the power series $P_{X+Y}(s)$ is

$$\sum_{k=0}^{m+n} P_X(k)P_Y(m+n-k)$$

which is as expected. We call C_k the convolution of the sequences A_K and B_k as the power series wherein the coefficient of x^i in C_k is

$$\sum_{k=0}^{i} a_k b_{i-k}$$

and we denote C_k as $A_k * B_k$. If we have three sequences $\{a_k\}, \{b_k\}, \{c_k\}$, then we will see that the convolution operation is commutative on all three of the sequences. Thus

$$\{a_k\} * \{b_k\} = \{b_k\} * \{a_k\}$$

and

$$\{a_k\} * (\{b_k\} * \{c_k\}) = (\{a_k\} * \{b_k\}) * \{c_k\}$$

Example 1: Let us find the probability generating function $\text{Bin}(n, p)$. Firstly, the probability generating function for a Bernoulli random variable with parameter p is $(1 - p + px)$. Therefore, the probability generating fucntion for the Binomial Random Variable is $(1 - p + px)^n$ Similarly, one can see that if X $\text{Bin}(n, p)$ and Y $\text{Bin}(m, p)$ then $X + Y$ $\text{Bin}(m + n, p)$

Exercise 1 : Find the probability generating function of $\text{Pois}(\lambda)$ and use it to find the sum of independent Poisson random variables.
Exercise 2: Find the generating function of geometric random variables.
Exercise 3: Find the probability generating function of negative binomial random variables.

Class 8, January 23 2026

Applications of Generating Functions

We have a sequence of i.i.d random variables such that

$$X_i = \begin{cases} +1, \text{ w.p } p \\ -1, \text{ w.p } q = 1 - p \end{cases}$$

And define $S_n = X_1 + \cdots + X_n$. Let us consider the event

$$\{S_1 \leq 0, \ldots, S_{n-1} \leq 0, S_n = 1\}$$

Let Φ_n denote the probability of the above event, and consider

$$\Phi(s) = \sum_{n \geq 0} \Phi_n s^n$$

If the above event occurs for some event then $S_1 = -1$. Let $\nu < n$ be the first time that this biased walk returns to the origin. Then our event is

$$\{S_1 = 0, S_2 \leq -1, \ldots, S_{\nu-1} = -1, S_\nu = 0\}$$

and the rest of the walk is such that

$$\{S_\nu \leq 0, \ldots, S_n = 1\}$$

This section of the walk depends only on the R.V.s $X_{\nu+1}, \ldots, X_n$. Therefore,

$$\Phi_n = \sum_{\nu=2}^{n-1} \mathbb{P}(S_1 \leq 0, \ldots, S_\nu = 0)\mathbb{P}(S_{\nu+1} \leq 0, \ldots, S_n = 1)$$

$$= \sum_{\nu=2} q\mathbb{P}_{-1}(S_1 \leq -1, \ldots, S_{\nu-1} = 0)\mathbb{P}(S_\nu \leq 0, \ldots, S_{n-\nu} = 1)$$

From this we get

$$\Phi_n = \sum_{\nu=2}^{n-1} q\Phi_{\nu-1}\Phi_{n-\nu}$$

Thus $\Phi_n = q(\Phi * \Phi)_{n-1}$. As for the generating function,

$$\Phi(s) = ps + \sum_{n \geq 2} \Phi_s s^n = ps + qs \sum_{n \geq 2} \Phi_n s^n = ps + qs(\Phi(s))^2$$

Therefore,

$$(\Phi(s))^2 qs - \Phi(s) + ps = 0$$

since this is a quadratic in $\Phi(s)$, we can solve for it to get

$$\Phi(s) = \frac{1 \pm \sqrt{1 - 4pqs^2}}{2qs}$$

But we know that $\Phi(0) = 0$ and by taking the limit as s goes to 0, we find that this generating function must actually be

$$\frac{1 - \sqrt{1 - 4pqs^2}}{2qs}$$

which gives us that

$$\Phi_{2k-1} = \frac{(-1)^{k-1}}{2q} \binom{\frac{1}{2}}{k} (4pq)^k$$

and

$$\Phi_{2k} = 0$$

As a result of this,

$$\Phi(1) = \sum_{n \geq 0} \Phi_n = \frac{1 - \sqrt{1 - 4pq}}{2q} = \frac{1 - |p - q|}{2q}$$

If $p \geq q$, then

$$\Phi(1) = \frac{1 - (p - q)}{2q} = 1$$

on the other hand, if $p < q$, then we get

$$\Phi(1) = \frac{1 - q + p}{2q} = \frac{p}{q}$$

So, if $p < q$, then

$$\mathbb{P}(\text{ walk never hits 1 }) = 1 - \frac{p}{q}$$

and if $p = q = \frac{1}{2}$, then

$$\mathbb{P}(\text{ walk eventually returns to 1 }) = 1$$

This suggests that if you are gambling with your friend by tossing a fair coin, then eventually you'd be filthy rich. Right?
Well, here's the thing. We must find the expected time that the walk hits 1. Let us generalize; we consider p and q as defined above. Then the expected time of hitting 1 is $1/(p - q)$ if $p > q$. But in the case that $p = q = \frac{1}{2}$, the expected time blows up to infinity. So expect to wait for a very long time or rather get hold of infinite capital. But then again, why would you gamble if you had so much money?

Class 9, January 30 2026

Simple Random Walk

We consider a simple random walk on $\mathbb{Z}^d$; ie. the d-dimensional lattice. We have a total of $2d$ choices at each step (two along each axis). We want to find

the probability that the walk returns to the origin. As in the one-dimensional random walk we consider u_{2n} to be the probability that the walk returns to the origin at the $2n$-th step and let f_{2n} be the probability that the walk returns to the origin for the first time at the $2n$-th step. Therefore, again as before we have:

$$u_{2n} = \sum_{k=1}^{n} u_{2n-2k} f_{2k}$$

and letting

$$U(s) = \sum_{n=0}^{\infty} u_{2n} s^{2n}$$

and

$$F(s) = \sum_{n=0}^{\infty} f_{2n} s^{2n}$$

we get that

$$U(s) = 1 + \sum_{n=1}^{\infty} \big(\sum_{k=1}^{n} u_{2n-2k} f_{2k} \big) s^{2n}$$

which simplifies to:

$$U(s) = \frac{1}{1 - F(s)}$$

We note that $F(1)$ is the probability that we eventually return to the origin. Due to this, we can say the following:

$$\lim_{s \to 1} U(s) = \infty$$

iff $F(1) = 1$. But $U(1) = \sum_{n=1}^{\infty} u_{2n}$. It would be enough to compute u_{2n}.

When $d = 2$, we choose the positions where we are going to move along the x-axis. and then we count the number of paths along the x-axis and the rest will be along the y-axis. From this we get

$$u_{2n} = \frac{\sum_{k=1}^{n} \binom{2n}{2k} \binom{2k}{k} \binom{2n-2k}{n-k}}{4^{2n}}$$

which comes out to be

$$\frac{\binom{2n}{n}^2}{4^{2n}}$$

Using Stirling's Approximation we find that $u_{2n} \sim \frac{1}{\pi n}$. Thus, this series diverges. And hence $F(1) = 1$ and thus the random walk is recurrent in 2 dimensions.

For $d = 3$, we consider a similar setting wherein we move along 3 axes now. For this we have:

$$u_{2n} = \binom{2n}{n} 2^{-2n} \sum_{j+k \leq n} \big(\frac{n!}{j!k!(n-j-k)!} 3^{-n} \big)^2$$

$$\leq \binom{2n}{n} 2^{-2n} \Big(\max_{j+k \leq n} \frac{n!}{j!k!(n-j-k)!} 3^{-n} \Big) \sum_{j,k} \frac{n!}{j!k!(n-j-k)!} 3^{-n}$$

We can check that the max is attained for $j, k \approx \frac{n}{3}$. This finally gives us that the random walk is transient in 3 dimensions.

Theorem 3 (Polya's Theorem)**.** *SRW is recurrent for $d = 2$ and transient for $d \geq 3$.*

Exercise: Verify Polya's Theorem using a neat argument.

Class 10, February 6 2026

Continuity Theorem

For each n, the sequence $a_{0,n}, a_{1,n}, \ldots$ is a probability distribution and thus $a_{k,n} \geq 0$ and

$$\sum_{k=0}^{\infty} a_{k,n} = 1$$

Theorem 4. *$a_k = \lim_{n\to\infty} a_{k,n}$ exists for all $k \geq 0 \iff$ The limit $A(s) = \lim_{n\to\infty} \sum_{k=0}^{\infty} a_{k,n} s^k$ exists for each $0 < s < 1$.*

Proof. $A_n(s) = \sum_{k=0}^{\infty} a_{k,n} s^k$. Assume the first statement. We want to show that $\lim_{n\to\infty} A_n(s)$ exists. Now

$$A_n(s) = \sum_{k=0}^{N} a_{k,n} s^k + \sum_{k=N+1}^{\infty} a_{k,n} s^k$$

and

$$\sum_{k=0}^{N} a_{k,n} s^k \leq A_n(s) \leq \sum_{k=0}^{N} a_{k,n} s^k + \frac{s^{N+1}}{1-s}$$

$$\sum_{k=0}^{N} a_{k,n} s^k \leq \liminf n \to \infty A_n(s) \leq \limsup_{n\to\infty} A_n(s)$$

Choosing N, large enough, we are done.
Next, we assume the second statement. Let us show that $a_{0,n}$ converges. Firstly, we note that $A(s)$ is monotonic in s and $A(0) = \lim_{s\to 0} A(s)$. Furthermore,

$$a_{0,n} \leq A_n(s) \leq a_{0,n} + \frac{s}{1-s}$$

and

$$A_n(s) - \frac{s}{1-s} \leq a_{0,n} \leq A_n(s)$$

Now,

$$A(s) - \frac{s}{1-s} \leq \liminf_{n\to\infty} a_{0,n} \leq \limsup_{n\to\infty} a_{0,n} \leq A(s)$$

Letting $s \to 0$, we get that LHS and RHS are the same. Therefore, $\lim_{n\to\infty} a_{0,n}$ exists. Let

$$B_n(s) = \frac{A_n(s) - a_{0,,n}}{s} = \sum_{k=1}^{\infty} a_{k,n} s^{k-1}$$

And we can carry out the same procedure as before to get the similar result for $a_{1,n}$. □

Some consequences of the continuity theorem

We have a sequence of random variables $\{X_n\}$ such that X_n $\text{Bin}(n, p_n)$ and if $np_n \to \lambda$ then

$$\mathbb{P}(X_n = k) \to \mathbb{P}(\text{Pois}(\lambda) = k)$$

Now the generating function for X_n is

$$[1 + p_n(1-s)]^n = [1 + \frac{np_n(1-s)}{n}]^n \to e^{\lambda(s-1)}$$

Next we consider X_n $\text{NB}(r_n, p_n)$. Suppose $p_n \to 1$, $r_n \to \infty$ such that $r_n(1 - p_n) \to \lambda$. Then

$$\mathbb{P}(X_n = k) \to \mathbb{P}(\text{Pois}(\lambda) = k)$$

Proof. We know that the generating function for X_n is

$$\left(\frac{p_n}{1-(1-p_n)s}\right)^{r_n}$$

We can write this as

$$e^{r_n[\log(1-(1-p_n)) - \log(1-(1-p_n)s)]}$$

Applying, the Taylor expansion for $\log(1-x)$ we get our desired result.

□

Class 11, February 12 2026

We consider a random walk starting at z with probability p of going to the right and q of going to the left. The gambler would win if he hits a but he loses if he hits zero first. We want to find the probability that the gambler hits 0 before hitting a. Let q_z be the probability for that event and let p_z be the probability that he hits a before hitting 0. We condition on the first jump.

$$q_z = pq_{z+1} + qq_{z-1}$$

with the initial conditions that $q_0 = 1$ and $q_a = 1$. We find that

$$q(q_z - q_{z-1}) = p(q_{z+1} - q_z)$$

$$\implies q_a - q_0 = \sum_{i=1}^{a} \left(\frac{q}{p}\right)^{i-1} (q_1 - q_0)$$

$$\implies -1 = \begin{cases} a(q_1 - q_0), \text{ if } p = q = \frac{1}{2} \\ \frac{\left(\frac{q}{p}\right)^a - 1}{\frac{q}{p} - 1}(q_1 - q_0), \text{ otherwise} \end{cases}$$

Thus

$$q_z = \begin{cases} 1 - \frac{z}{a}, \text{ if } p = q = \frac{1}{2} \\ \frac{\left(\frac{q}{p}\right)^a - \left(\frac{q}{p}\right)^z}{\left(\frac{q}{p}\right)^a - 1}, \text{ otherwise} \end{cases}$$

Due to symmetry, one could flip the roles for p and q to get that $p_z + q_z = 1$. After $a - 1$ steps, $\mathbb{P}(\text{walk is out of interval}) \geq p^{a-1}$. If we're out, then good. If not, consider next $a - 1$ steps. Again we have the same probability of getting out. Thus intuitively, we have infinitely many attempts to make $a - 1$ moves. Thus the walk will eventually get out. If we have the probability for success being p and that of failure being p, then the probability that we fail k-times before succeeding is $q^k p$. Thus we probability that we fail forever is

$$1 - p \sum_{k=0}^{\infty} q^k = 0$$

So we must eventually succeed. When $a \to \infty$, we have that $q_z \to 1$. But when $q < p$, there is a positive chance we may never hit 0.